

A brief overview of Deep Unfolded Fractional Programming-Based Beamforming in RIS-Aided MISO Systems

Jannes Michael Plett

I. SYSTEM MODEL

A. General Data

- BS has M antennas
- K user
- RIS with N elements
- C. BS and user k : $\mathbf{h}_{d,k} \in \mathbb{C}^M$
- C. RIS and user k : $\mathbf{h}_{r,k} \in \mathbb{C}^N$
- C. BS and RIS: $\mathbf{G} \in \mathbb{C}^{N \times M}$
- Phase Shift of RIS: $\boldsymbol{\theta}$

The user receives from BS:

$$y_k = (\mathbf{h}_{d,k}^H + \mathbf{h}_{r,k}^H \text{diag}(\boldsymbol{\theta}) \mathbf{G}) \sum_{k=1}^K \mathbf{v}_k s_k + n_k \quad (1)$$

II. THE DESIGN, INTENT, AND LIMITATIONS OF THE TEMPLATES